Since COM-HPC server-on-modules are generally not implemented in dual-processor boards, the standard positions itself in the mid-performance segment of server technology, which is required in many diverse configurations at the edge.

## 5G infrastructure servers and connected edge systems

Much of the demand for robust server-class performance comes from edge applications that reside directly in and at the network and communication infrastructures. Examples are outdoor small cells on 5G radio masts for tactile broadband Internet, or carrier-grade edge server designs with a 50.8cm (20-inch) installation depth, which are currently found in 3G/4G edge server rooms where the installation depth of racks is just 600mm. In both the cases, the systems are often enriched with GPU or FPGA boards for future 5G designs. COM-HPC server modules are, of course, also attractive for the related 5G measurement technology.

## OCCERA server with full NFV and SDN support

Another application area for COM-HPC server modules is robust edge server designs that are based on the Open Compute Carrier-grade Edge Reference Architecture (OCCERA) and built-in rack depths of 1000mm or 1200mm. Due to increasing virtualisation and support for NFV and SDN, dedicated infrastructure equipment is being implemented on such open standard platforms in growing numbers. This includes firewalls, intrusion detection systems, traffic shaping, content filters, deep packet inspection, unified threat management, and antivirus applications. Here too, users benefit from on-demand scalability and cost-effective performance upgrades for the next generation of servers.

## From Industry 4.0 to workload consolidation

Ultimately, edge equipment also includes all the fog computers close to the applications, that is, redundant clouds at the edge. In the harsh industrial environment, these are required for Industry 4.0 communication and/or to implement new business models and comprehensive predictive maintenance

| COM-HPC™ Server | COM Express Type 7 |
|---|---|
| 65x PCIe | 32x PCIe (Gen 3.0) |
| 8x 25GbE KR | 4x 10GbE KR |
| BaseT (up to 10 Gb) | 1x 1GbE |
| 2x USB 4 | 4x USB 3.0 |
| 2x USB 3.2 | 4x USB 2.0 |
| 4x USB 2.0 | 2x UART / CAN |
| 2x SATA | 2x SATA |
| eSPI, 2x SPI, SMB | LPC / eSPI |
| 2x I2C, 2x UART | I2C & SPI |
| 12x GPIO | 8x GPIO/SDIO |

*Fig. 2: COM-HPC server offers significantly more interfaces than COM Express Type 7 modules*

using Big Data analytics at the edge. Another application area is opening up the possibility to consolidate the computing performance on a single edge server instead of distributed computers, for instance, to control a manufacturing cell with several robots.

## Energy grids, autonomous vehicles, and rail transport

Similar edge servers are also needed in smart energy networks to reconcile energy generation and consumption in microgrids and virtual energy networks in real time. In such distributed applications, real-time 5G plays a great role since the required high bandwidths along the infrastructure paths can be provisioned easily.

Autonomous vehicles will benefit from 5G, too. These also need COM-HPC server performance in the cockpit as these have to process high bandwidths onboard to analyse data from hundreds of sensors in fractions of a second. Further applications can be found in transportation, namely in the rail sector, to provide a full range of services—from security technology to streaming services for infotainment. Mobile broadcasting systems and streaming servers, as well as equipment for non-public mobile terrestrial radio networks for use by security authorities and organisations, are further application areas for COM-HPC server modules.

## Smart vision systems

A considerable number of COM-HPC server applications can also be found in medical backend systems, industrial inspection systems, and video surveillance solutions at the edge for increased public safety, whereas with autonomous vehicles, massive video and imaging data have to be processed in parallel. However, a significant share of these applications will not rely on COM-HPC GPGPU but on using even more powerful FPGA extensions or PEG graphics.

## Modular motherboard designs

In addition to the often dedicated system designs for the application areas described so far, COM-HPC modules will also be deployed on robust standard motherboards, which are long-term available and ideal for the design of smaller servers. Used mostly in rackmount, workstation, or tower designs, these boards range from the µATX format (244mm × 244mm) to the extended ATX form factor (305mm × 244mm). EEB mainboards (305mm × 330mm) will also be able to use COM-HPC.

As some of the latter feature two processors, these will only be able to conquer part of this form factor market. However, these will bring previously unimaginable scalability to this market, coupled with the option to customise the I/O interfaces extremely cost-effectively and efficiently. This is an enormous advantage for edge applications as the industrial communication landscape is extremely heterogeneous in the direction of the process and field level. So, in summary, COM-HPC server modules are suitable for an extremely wide range of edge and fog server applications in harsh environments, whereby the focus tends to be clearly on the processor.

## Why use COM-HPC server modules?

The COM-HPC server specification offers edge servers the full advantages of standardised server-on-modules, which we are familiar with from the computer-on-module market, that is, the performance can be tailored to the specific needs of the application. In a rack with identical server slots, it is therefore possible to adapt the physical performance through the design of each individual module, thereby opti-